MILITARY SPECIFICATION

RADIO SET, AN/ARC-39

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide voice amplitude modulated communication between helicopter type aircraft and ground stations over the frequency range of 2.0 to 9.1 mc/s.

1.2 Classification - The Radio Set AN/ARC-39 shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size (Inches) W	H/L	D	Max. Allow. Wt. Lbs.	Appl. Para.
Receiver-Transmitter	RT-427/ARC-39	B1 - D1			36.25	3.5.1
Control, Radio Set	C-2241/ARC-39	5-3/4	2-1/4	2-1/8	1.0	3.5.2
Mounting						
Mounting Base	MT-605/APR-9	MT-B1 - D1			2.75	3.5.1.2

1.3 Accessories - The equipment shall operate with the following accessories which are not to be supplied as part of this equipment.

(1) Antenna - Fixed or trailing wire

(2) Headset - H-1/AR or equal

(3) Microphones - M-3A/UR, Boom Assembly, ASO Stock No. R16M2350 (Hand Held Carbon) or equivalents.

(4) Throttle Switch - NAF-1124-7

(5) Interphone - AN/AIC-4 or equal

(6) Range Receiver - R-23A/ARC-5

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-1	Electron Tubes and Crystal Rectifiers
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-C-3098	Crystal Units; Quartz
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft, Characteristics of
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

Bureau of Aeronautics

EI-207	Control Panels, Electrical and Electronic Equipment, Rack or Console Mounted, Installation of
SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins, Preparation of

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STANDARDS

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Air Force - Navy Aeronautical

AND 10461	Connections - Input Power Airborne Electronic Equipment; 28 Volt DC, 115 Volt Single Phase AC, and/or 115/200 Volt Three Phase AC

2.2 Availability of Documents:

(1) Give both the title and number when requesting specifications, publications and drawings. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials as approved by specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard MIL parts or materials will not fulfill the design objectives because of size, weight, performance, or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions, as required by specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

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(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the procuring agency.

3.1.1 Non-Standard Part and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of specification MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in specification MIL-E-5400, of an electrical nature shall be approved before the preproduction model of the equipment is delivered to the Bureau of Aeronautics. The request for approval shall be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Choice of Tube Types - Electron tubes used in the equipment shall be chosen in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor must submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

3.2 Design and Construction

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of specification MIL-E-5400, unless otherwise specified herein.

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3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 25°C (+15°C or -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in specification MIL-E-5400, except as follows:

3.2.3.1 Altitude - The altitude requirement of 50,000 feet specified in specification MIL-E-5400 shall be reduced to 25,000 feet.

3.2.4 Total Weight - The total weight of the equipment excluding cables, shall be a minimum consistent with good design and shall not exceed 40 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 100 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Console Controls - All remote control units shall be satisfactory for mounting in a console arrangement when used in aircraft. These units shall meet the requirements of specification MIL-C-6781, shall have a plastic lighting plate meeting the requirements of specification MIL-P-7788 and shall be equipped with fasteners meeting the requirements of specification MIL-F-25173 for mounting in accordance with requirements of specification EI-207. The lamps shall be connected in parallel, with one side grounded to the chassis of the unit and the other side being connected to a separate contact, in the electrical connector receptacle of the unit, for connection to an external +28 V input lead. The brilliance of the lamps shall be controlled externally and the brilliance control shall not be part of the control unit. The arrangement of controls and all lettering and markings proposed for use on the front panel of control units shall be approved by the Bureau of Aeronautics prior to fabrication of any complete units.

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3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in specification MIL-E-7894 having limits as specified herein. The power required shall not exceed the following:

(1) DC Power 28 Volts System

Operating Voltage Limits	25 to 29 Volts
Reduced Performance Voltage Limits	20 to 25 Volts
Current.	9 Amps

3.2.11 Input Power Connection - Input power connections shall be in accordance with drawing AND 10461.

3.2.12 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.12.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Moisture and Fungus Resistant Treatment - Equipments shall be fungus-proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.14 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with the requirements of specification MIL-N-18307.

3.2.15 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in specification MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of specification MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.16 Provisions for Maintenance and Field Testing - The equipment shall provide for ease of maintenance to the greatest extent practicable. Provisions for field testing shall be as specified in specification MIL-E-5400.

3.2.17 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of specification MIL-I-6181, except as follows:

3.2.17.1 Conducted Interference - The requirements of specification MIL-I-6181 regarding conducted interference as applied to transmitter key-down conditions shall be waived insofar as transmitter fundamental frequency interference is concerned, and shall be relaxed to allow, at harmonics of the transmitter frequency, a maximum of 300 microvolts conducted interference on any external conductor which could conduct interference to other equipment.

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3.2.17.2 Radiated Interference - The radiated interference requirements of specification MIL-I-6181 shall not apply under transmitter key-down conditions.

3.2.17.3 Susceptibility Limits, Radio Frequency - The RF susceptibility limits of specification MIL-I-6181 shall apply except as follows: The signal level required to produce a 6 db signal-plus-noise-to-noise ratio, when applied to the power line shall be 60 db greater than the signal level required at the receiver antenna terminals to produce the same signal-plus-noise-to-noise ratio. The signal levels required above shall be supplied by a signal generator (with 50 ohms source impedance) which is tuned to the receiver operating frequency. With the signal generator adjusted to the intermediate frequency of the receiver, the signal level required to produce a 6 db signal-plus-noise-to-noise ratio when applied to the power line shall be greater than 10,000 microvolts.

3.2.17.4 Susceptibility Limits, Audio Frequency - The AF susceptibility limits of specification MIL-I-6181 shall apply except as follows:

3.2.17.4.1 The receiver audio output, due to superimposing a sine wave audio-frequency signal of 2 volts rms on the 28 volt DC power line shall be measured with a 1000 microvolt unmodulated signal applied to the receiver input terminals. When the frequency of the audio signal on the power line is adjusted to values between 200 and 10,000 cps, the receiver audio output due to this signal shall be less than that which results from modulating the signal input to a level of 3% at 1000 cps. When the frequency of the audio signal on the power line is adjusted to values between 60 and 200 cps, the receiver audio output due to this signal shall be less than that which results from modulating the signal input to a level of 10% at 1000 cps.

3.2.17.4.2 Modulation of the transmitter output due to superimposing a sine wave audio-frequency signal of 2 volts rms on the 28 volts DC power line shall not exceed 3% for power line signal frequencies between 200 and 10,000 cps or 10% for power line signal frequencies between 60 and 200 cps. This performance shall be obtained with the clipping level control adjusted to provide 95% modulation, unclipped, with a 0.18 volt (open circuit), 1000 cps signal supplied by the dummy microphone.

3.3 Performance - The equipment shall meet the following performance requirements:

3.3.1 Warm-Up Time - The equipment shall operate normally within 1 minute after turn on except that 15 minutes shall be allowed for stabilization of frequency.

3.3.2 Frequency Characteristics - Without the use of any type of tools or test equipment, it shall be possible to conduct all operations necessary for adjustment of the equipment to 12 preset crystal-controlled frequencies with 6 such preset channels located anywhere in the band of 2.0 to 4.27 mc/s and the remaining 6 located anywhere in the band of 4.27 to 9.1 mc/s. It shall be possible to select any one of the twelve preset frequencies by means of front panel controls on either receiver-transmitter, RT-427/ARC-39 or radio set control, C-2241/ARC-39. Voice transmission and reception shall be available on any preset frequency.

3.3.3 Modulation Characteristics

3.3.3.1 Capability - The modulator of this equipment shall be capable of symmetrically modulating the RF carrier to a level of 95 to 100% when a 1000 cps input voltage from the dummy microphone (see paragraph 3.3.20.2) is adjusted to 1.0 volt open circuit. A clipper circuit shall be incorporated in the modulator and shall symmetrically clip both positive and negative peaks of the modulating voltage. There shall be 2 controls provided whereby, with a 1000 cps input voltage of 1.0 volt open circuit from the dummy mic. phone, the percentage modulation at which clipping just starts can be adjusted between 60% and 95% and whereby the degree of clipping available can be adjusted between 0 and 15 db.

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3.3.3.2 Fidelity - The modulation output relative to that which produces 90% modulation at 1000 cps shall be down not more than 6 db from 300 to 1000 cps, nor more than 3 db from 1000 to 3500 cps. Total harmonic distortion at 90% modulation with 1000 cps (without clipping) shall not exceed 10%. Carrier noise level shall be at least 40 db below the RF output level with 90% modulation.

3.3.3.3 Sidetone - The sidetone output when the transmitter carrier is modulated 30% at 1000 cps shall be adjustable from a maximum greater than 50 mw to a minimum less than 1 mw. Sidetone shall be obtained from rectification of the modulated carrier.

3.3.3.4 Microphone Current - The carbon microphone energizing current shall be 50 milliamperes ± 5 milliamperes when measured with the dummy microphone connected and with the equipment operated from the power source, specified in paragraph 3.2.10, adjusted to 27.5 volts.

3.3.3.5 Filtering - Suitable filtering shall be employed to attenuate harmonics of the clipped speech above 3500 cps.

3.3.4 Frequency Stability - After a warm-up period of 15 minutes, the frequency stability of the transmitter shall be such that the frequency excursion over a full range of service conditions shall not exceed ± 0.01% of the assigned frequency. This stability tolerance is overall and shall include all frequency changes accompanying operation of the equipment under service conditions, plus frequency changes due to the inaccuracies in the channel selecting mechanism.

3.3.5 Antenna Circuits - The transmitter antenna tuning circuits shall be capable of resonance, and proper loading when connected to antennas having effective capacitances and resistances, as seen at the antenna binding post, in accordance with the following:

Frequency (Mc/s)	Capacitance (mmf)	Resistance (ohms)
2.0	60 to 300	0 to 5
3.0	30 to 300	0 to 5
6.0	30 to infinity	0 to 5
9.1	30 to infinity	0 to 5

3.3.5.1 A meter circuit shall be provided to indicate proper antenna resonance and power amplifier loading

3.3.6 RF Power Output - When connected to a 10 ohm 100 micromicrofarad dummy antenna, the transmitter shall be capable of delivering at least the following values of current to the load.

Frequency(Mc/s)	Antenna Current (Amps)
2.0	1.0
3.0	1.1
5.0	1.1
9.1	1.1

3.3.6.1 When the transmitter is connected to a 0 ohm, 60 mmf antenna at 2 mc/s, the antenna current shall be at least 1.4 amperes.

3.3.7 Spurious Radiation - Harmonics or other spurious radiation shall be effectively suppressed from the RF output of the transmitter. The signal strength of any radiated signal on other than carrier frequency shall be 50 db or more below the carrier signal.

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3.3.8 Noise Output - Modulation of the RF carrier, emitted by the transmitter and attributable to vibration, shall be 40 db or more below the 90% modulation level.

3.3.9 Channel Changing Time - The time elapsing, between the instant the channel selector switch is in position to select a given channel and the instant the equipment is in an operative condition on the selected channel, shall be not greater than 6 seconds under any service conditions.

3.3.10 Sensitivity - With an RF input of 5.0 microvolts modulated 30% at 1000 cps, the receiver audio output into a 300 ohm non-inductive resistive load shall be at least 40 mw, and the signal-plus-noise-to-noise ratio (modulation ON vs. modulation OFF) shall be at least 6 db. The sensitivity measured under service conditions shall not be degraded by more than 4 db from that measured under standard conditions.

3.3.11 AGC Characteristics - The AGC action shall be such that, when the receiver output level control is set for maximum output, and when a 1000 microvolt signal, modulated 30% at 1000 cps, is applied, the audio power output shall be 200 mw (-25, +50%). When the modulated signal level is varied from 10 to 100,000 microvolts the audio power shall not vary by more than 6 db, and when the modulated signal level is raised to 2 volts, the audio output power shall not decrease by more than 6 db from maximum output.

3.3.12 Selectivity - The total bandwith at 6 db down from normal sensitivity at the resonant frequency shall be 8.0 kc/s (± 1 kc/s) and at 60 db down shall be not greater than 3.5 times the 6 db bandwidth. The center frequency of the receiver bandpass as determined from the 6 db attenuation points relative to maximum response, shall not vary more than ± 0.01% ± 2.0 kc/s from the assigned frequency. The response curve shall be essentially symmetrical about the resonant frequency.

3.3.13 Image Frequency Rejection - The receiver shall provide an attenuation of not less than 85 db, relative to the output at the resonant frequency, to all images and other spurious responses resulting from the combination of fundamental or harmonic frequencies of the conversion oscillator with undesired signals.

3.3.14 IF Rejection - The receiver shall provide an attenuation of not less than 85 db, relative to the output at the resonant frequency, to signals of the frequency of the intermediate amplifier.

3.3.15 Regeneration - Oscillation or instability due to regeneration shall not be present in the receiver when the RF gain is maximum, a 30 volt power source is applied, and all vacuum tubes used have transconductances in the upper 50% of the range specified in the applicable tube specification sheets of specification MIL-E-1. The method of checking regeneration shall be subject to the approval of the Bureau of Aeronautics.

3.3.16 Noise Peak Limiter - The second detector of the receiver (that which follows the intermediate frequency amplifier) shall be provided with an instantaneous peak limiter. The noise limiting action shall effectively limit or reduce voltage surges or peaks exceeding the receiver output obtained with a 65 to 85% modulated signal at maximum sensitivity. Its action shall be relatively instantaneous so as not to obliterate signals by a permanent reduction in receiver gain under the influence of continuous RF noises such as are caused by commutators, ignition systems, etc. When such noise is introduced into the receiver input circuits in conjunction with a sinusoidally modulated signal having an amplitude of modulation corresponding to the threshold of noise limiter action, the peak amplitude of the audio output, when viewed with an oscilloscope, shall not exceed the peak amplitude without noise by more than 50%.

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3.3.17 Audio Characteristics - The receiver audio output distortion shall be less than 10% with an RF input of 10 to 100,000 microvolts, modulated 30% at 1000 cps. The audio output for 30% modulated signals relative to that obtained for 1000 cps modulation shall not be down more than 3 db for modulation frequencies from 300 to 2500 cps and shall not be down more than 6 db at 3500 cps. The maximum practicable attenuation shall be attained below 300 cps and above 3500 cps without recourse to inclusion of large and/or heavy filter components in the receiver.

3.3.18 Spurious Radiation - The maximum radio frequency voltage, due to the receiver oscillator, appearing across a 100 micromicrofarad capacitor connected between the antenna post and the front panel shall be less than 50 microvolts.

3.3.19 Volume Control - The output level control shall have a range sufficient to attenuate the audio level at least 35 db from the maximum level attainable. This control shall be approximately linear over the range of the control.

3.3.20 Audio Terminations for Test Purposes -

3.3.20.1 Receiver - The audio output channel of the receiver shall be terminated in a 300 ohm non-inductive load which shall include means suitable for measuring the audio frequency power output.

3.3.20.2 Transmitter - The audio input circuit of the transmitter shall have connected thereto a dummy microphone having the following characteristics:

Internal Resistance ----------82 ohms
Impedance (mainly resistive at AF) ----------82 ohms
Microphone Circuit ---------- Unbalanced, one terminal grounded
Frequency Range ----------150 to 10,000 cps
Total Harmonic Distortion ----------Less than 5%
Open-circuit Output Potential----------0.01 to 3.0 volts rms (adjustable)
Meter, AC (to indicate open-circuit voltage regardless of load) -- 0 to 3.0 volts rms
Meter, DC ----------0 to 100 ma

3.4 Workmanship - Workmanship, fabrication, and manufacturing processes shall conform to the requirements of specification MIL-E-5400.

3.5 Detail Requirements -

3.5.1 Receiver-Transmitter, RT-427/ARC-39 - The receiver-transmitter unit shall be designed to meet the following requirements:

3.5.1.1 Function - The transmitter section of the receiver-transmitter unit shall generate RF power and transmit it to the antenna system. The receiver section of the receiver-transmitter shall receive RF energy from the antenna system, amplify it, and convert it to aural intelligence.

3.5.1.2 Form Factor and Mountings - The overall dimensions of the receiver-transmitter unit shall not exceed those specified in specification MIL-C-172 for case size B1-D1 and utilizing the maximum case extension and forward projection indicated therein. The receiver-transmitter mounting MT-605/APR-9 shall be suitable for attachment to a plane surface in an aircraft and for supporting the receiver-transmitter unit, which shall be removable without the use of tools. The mounting shall be similar to the MT-B1-D1 size as per specification MIL-C-172.

3.5.1.3 Weight with Mountings - The weight of the receiver-transmitter unit including the mounting, with vibration isolators attached shall not exceed 39 pounds.

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3.5.1.4 Contents of Unit - The unit shall contain the transmitter, receiver, and dynamotor power supply. All semi-fixed adjustable devices provided for alignment, gain, etc. shall be readily accessible when the receiver-transmitter is withdrawn from its case. If tools other than a conventional straight-tipped screwdriver and conventional wrenches for locknuts are required for the manipulation of these devices, they shall be furnished and attached to the receiver-transmitter chassis. The receiver-transmitter shall contain the following assemblies:

3.5.1.4.1 Main Chassis Assembly - The main chassis assembly shall constitute the framework including the front panel, to which all other assemblies are secured. It shall include the transmitter modulator circuits, the ganged tuning capacitors for tuning both receiver and transmitter, and a panel meter with switching for use in transmitter tune-up. Power input connections shall be made to a receptacle mounted on the front panel.

3.5.1.4.2 Transmitter RF Assembly - The transmitter RF assembly shall include the tubes and circuits used to generate and amplify the transmitter frequency. It shall also include the receiver oscillator.

3.5.1.4.3 Antenna Tuner Assembly - The antenna tuner assembly shall include the network used to match the antenna to the output of the transmitter RF assembly.

3.5.1.4.4 Antenna Relay Assembly - The antenna relay assembly shall include the necessary switching facilities to transfer the antenna from receiver to transmitter, and a filter network to permit simultaneous use of the antenna by the receiver of the AN/ARC-39 and by a range receiver R-23A/ARC-5.

3.5.1.4.5 Preselector Assembly - The preselector assembly shall include the RF amplifier and mixer tubes and associated circuits for the receiver.

3.5.1.4.6 IF Audio Assembly - The IF audio assembly shall include the IF and audio amplifier tubes and associated circuits for the receiver. Some of these tubes shall also amplify the transmitter sidetone output.

3.5.1.4.7 Retuner Assembly - The retuner assembly shall be an electromechanical assembly which repositions each of four shafts to the selected one of twelve locations previously established during equipment tune-up. The retuner assembly is normally remotely controlled by means of the Radio Set Control C-2241/ARC-39, but may be controlled by means of a local selector switch located on the retuner assembly front panel. The retuner assembly includes the sockets and selector switch for the quartz crystals used to control the receiver-transmitter channel frequencies.

3.5.1.5 External Controls

3.5.1.5.1 Remotely Operated Controls - The following controls, required for tuning the equipment, shall be mounted on the front panel. All of these controls shall be actuated by electromechanical means to effect the 12 channel preset operation specified herein.

(1) Receiver and Transmitter Tuning (ganged)

(2) Power Amplifier Loading

(3) Transmitter Antenna Tuning, Coarse

(4) Transmitter Antenna Tuning, Fine

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3.5.1.5.2 Operating Controls and Metering - The following controls, used only during tune-up procedures, shall also be mounted on the front panel.

(1) Receiver Antenna Trimming, High Band

(2) Receiver Antenna Trimming, Low Band

(3) Local Channel Selector

(4) Local Audio Level

(5) Sidetone Level (with suitable lock)

(6) Meter Switch

(7) Meter, for measurement of power amplifier grid current and cathode current

3.5.1.6 Internal Controls - The following controls shall be located within the unit in such positions as to be accessible for adjustment with the case removed.

(1) Clipping Level Adjustment

(2) Modulation Level Adjustment

(3) All necessary RF and IF alignment adjustments

3.5.1.7 Electrical Connections - All electrical connections to the receiver-transmitter unit shall be made to suitable connectors mounted on the front panel. All power input, control and audio connections necessary for operation shall be made through a single multi-contact plug. Additionally, a headset phone jack and a microphone jack shall be provided on the front panel for use during tune-up procedures. Two binding posts shall be provided, one for connection to the antenna and the other for connection to an auxiliary range receiver for operation on the same antenna. Sockets for twelve receiver crystals and twelve corresponding transmitter crystals shall also be provided on the front panel under a removable protective cover.

3.5.2 Control, Radio Set C-2241/ARC-39 - The radio set control unit shall be of the console type and shall fulfill the following requirements:

3.5.2.1 Function - The radio set control shall provide remote control of the receiver-transmitter by the pilot.

3.5.2.2 Form Factor - All dimensions shall be in accordance with specification MIL-C-6781. The dimensions parallel to the mounting rails shall not exceed 2-1/4 inches.

3.5.2.3 Weight - The weight of the unit shall not exceed 1.0 pound.

3.5.2.4 Contents of Unit - The unit shall contain an ON-OFF switch combined with a rotary control for adjusting audio output level, a twelve-position rotary channel selector switch, and a switch for transferring channel selection control to or from the front panel control of the receiver-transmitter unit.

3.5.2.5 Controls - The following operating controls shall be located on the front panel of the unit:

(1) VOL-OFF control; turns power to equipment on and off, and controls audio output level.

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(2) CHANNEL selector switch; remotely selects one of 12 preset channels. A lever recessed in the knob used on this switch actuates a second switch which transfers channel selection control to or from the LOCAL SELECTOR switch on the front panel of the receiver-transmitter unit.

3.5.2.6 Electrical Connection - All electrical connections for control, audio, and lighting circuits shall be made to a single receptacle located on the rear of the unit.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests

(1) Preproduction Tests - Preproduction Tests are those tests conducted on an equipment representative of the equipments to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction Tests shall be accomplished by Contractor Demonstration Tests and Service Approval Tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests shall be accomplished under the responsibility of the contractor to demonstrate that the equipment complies with all applicable requirements.

(b) Service Approval Tests - Service Approval Tests shall be accomplished under the responsibility of the procuring service for the purpose of verifying the results of the Contractor Demonstration Tests and for conducting any additional tests deemed necessary to determine that the equipment complies with all applicable requirements.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on equipment submitted for acceptance under the contract. Acceptance Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics and which have previously passed Individual Tests.

(c) Special Tests - Special Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics.

(3) Life Tests - Life Tests shall be conducted by the contractor at his plant. These tests shall be started as soon as practicable on an early production equipment.

4.2 Preproduction Tests -

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more equipments as necessary to determine that the model as proposed by the contractor will meet all the requirements of the applicable specifications and the contract. Contractor Demonstration Tests shall be conducted in accordance with the approved test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the equipment for Service Approval Tests. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. When tests are supervised, the Bureau of Aeronautics may, when conducting Service Approval Tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting Service Approval Tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary. Contractors not having laboratory facilities to conduct all required tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

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cycles	repeat	cycles		to		percent	repeat	cycle	repeat

4.2.2 Service Approval Tests - The Service Approval Sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for Service Approval Tests, the contractor shall also submit the accessory material and design and test data specified in MIL-E-5400. The design and test data shall indicate the physical and electrical characteristics of the equipment and establish that the equipment meets all applicable requirements. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction Tests shall include environmental tests in accordance with MIL-T-5422 and interference tests in accordance with MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance under the contract. The preproduction models shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Bureau of Aeronautics shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipments as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests and has been packaged and packed ready for shipment.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as approved or requested by the Bureau of Aeronautics.

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4.3.1 Individual Test - Each equipment submitted for acceptance shall be subjected to the Individual Test to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall successfully pass the tests specified in the following paragraph.

4.3.1.1 Scope of Test - As a minimum, each equipment shall be subjected to the following:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1.1 Examination of Product - Each equipment shall be carefully examined to determine that the requirements for material and workmanship as specified in MIL-E-5400 have been met. Also, it shall be determined that approval has been granted for the use of all non-standard parts.

4.3.1.1.2 Operational Test - Each equipment shall be operated. The period of operation shall be long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.1.3 Reliability Test - Each equipment shall be operated (without isolators) under the conditions specified below for a period of 12 hours without a failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only such adjustments shall be permitted which can be made using such controls and adjustments which would normally be accessible during flight.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any selected frequency within the range of 10 to 30 cps (excluding resonant points) and an amplitude of ± 3 g's

The ten (10) minute vibration run shall be made at the beginning of the twelve (12) hour period. The equipment shall be nonoperating during the vibration run. During the twelve (12) hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except the ten (10) minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action must be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The twelve (12) hour period as specified above may be composed of several short periods of three (3) hours or more.

4.3.2 Sampling Test - Sampling Tests shall be conducted on a quantity of equipment as specified in the following paragraph. Equipments selected must first have passed all of the individual tests. Sampling Tests shall be conducted on complete equipments.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment. Selection of equipment for tests shall be under the supervision of the naval inspector.

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Quantity Delivered for Acceptance	Quantity to be tested
From the first 10	1
From 11 to 100	2
From 101 to 200	1
From 200 and above	1 for each additional 200 or fraction thereof

4.3.2.2. Scope of Test - As a minimum each equipment selected for the Sampling Test shall be subject to the tests specified below.

Tests	
I	Conduct a complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure all applicable specification requirements have been met.
II	Repeat the above using the highest and lowest specified input voltages and frequencies.
III	Conduct the reliability test specified in paragraph 4.3.1.1.3 except the test duration shall be 50 hours and two failures shall be permitted.

4.3.3 Special Tests - Special Tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. Equipment selected for Special Test may be selected from equipment previously subjected to the Sampling Test.

4.3.3.1 Special Test Schedule - In selecting equipment for Special Tests the following schedule shall be followed:

(1) An early production equipment.

(2) An equipment midway of the production run.

(3) An early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

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VARIABLE TRANSFORMER (SEE TABLE III)

4.3.3.2 Scope of Test - Special Tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the Preproduction and Sampling Tests shall be used where applicable. When not applicable the contractor shall prepare a test procedure and submit to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that, for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (45 minutes cycling - 15 min. in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the Sampling or Special Tests, the following action shall be taken.

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect.

(3) Suggest corrective action, for Bureau of Aeronautics' approval, designed to reduce the possibility of the same failure occuring in future tests.

(4) Where practical, include a test in the Individual Test to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Tests - The equipment shall be subjected to a Life Test at the contractor's plant under the following simulated service conditions.

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	26. 5 ± 2. 5 volts

4.4.1 Test Data - The contractor shall keep a daily record as to the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures during this test, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part Type number

(2) The circuit reference symbol-number

(3) The part function

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(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

In the event of a part failure the naval inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occured. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.2 Test Duration - The duration of the Life Test shall be 300 hours.

4.4.2.1 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically the equipment shall be turned on and off several times and put through all phases of its performance.

4.4.2.2 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.3 Life Test Schedule - The equipment to be tested shall be representative of the current production in all significant aspects of design and manufacture. It shall be selected by the Government Inspector from equipments which have passed the Individual Tests. The Life Test shall be performed on the following quantities of equipment unless otherwise stated in the contract or approved by the Bureau of Aeronautics.

Quantity Delivered for Acceptance	Quantity to be tested
First 25	1
Next 26 to 200	1
Next 201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.4 Reconditioning of Test Samples - Equipment which has been subjected to the Life Test shall be reconditioned as follows:

(1) On completion of the Life Test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring agency.

(2) After reworking, the contractor may submit the equipment for acceptance under the contract.

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4.5 Test Procedures - The procedures used for conducting Preproduction Tests, Acceptance Tests and Life Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts, the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements. The design data submitted by the contractor for the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This applies for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with the requirements of SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, pr ed and marked for shipment in accordance with the requirements of Specifications MIL-P-17555 and MIL-STD-129.

6. NOTES

6.1 Use - The primary use of this equipment is to provide two-way voice communications (A-3) in helicopter type aircraft.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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